

DSA Roadmap

Basic of java with dsa

1. Print Statement, If else Statement
2. Arrays, For loops
3. Modulus Operator, And Or (&, |) operators
4. Reversing a String, Palindrome, Boolean datatype, break
5. Function
6. Ascii values, type casting, adjacent
7. Leetcode problems arrays strings
8. Nested double for loop 2 leetcode problems
9. Patterns (very important to improve logic, double for loops)
10. Patterns - 2 (Very Important to Improve Logic, double for loops)
11. Time Complexity, Best time to Buy and Sell Stocks.
12. 3 leetcode problems | Arrays .
13. Inputs in Java
14. Bubble Sort | Array Sort
15. Hashmap in java
16. Hashmap part – 2, Leetcode problem
17. Leetcode problems on Hashmap
18. Leetcode problems on Hashmap urra mass problems
19. Leetcode problems on Hashset
20. 2D arrays
21. 2D arrays leetcode problem
22. ArrayList, Pascals Triangle Leetcode 🙋 (use the below link for continuation)

<https://www.vigneshreddyjulakanti.in/> - java basics to dsa roadmap

<https://takeuforward.org/dsa/strivers-a2z-sheet-learn-dsa-a-to-z> - Strivers DSA Sheet A2Z

<https://www.youtube.com/watch?v=VTLCoHnyACE&list=PLfqMhTWNBT137IEPQd34TsgV6IO55pt> - DSA with C++

<https://dsa.apnacollege.in/> - DSA sheet starting from arrays

<https://www.youtube.com/watch?v=yRpLIJmRo2w&list=PLfqMhTWNBT137IEPQd34TsgV6IO55pt> – DSA with java

<https://www.youtube.com/watch?v=bSrm9RXwBal&pp=ygUVYXBuYSBjb2xsZWdlIGRzYSBqYXZh> –
oops

<https://www.youtube.com/watch?v=SqcXvc3ZmRU&list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX>
– system design

<https://www.youtube.com/watch?v=Xx1eYBIUGO8&pp=ygUbc3lzdGVtIGRlc2lnbiBjb3Vyc2UgdGVsdWd1> – system design

WEEK 1

Day 1-2

DSA : Arrays basics – traversal,sum,max 2 simple problems

System Design – What is system design? And concepts like scalability latency, throughput

Day 3-4

DSA : Strings (reverse,palindrome)

System Design - Client–Server architecture and Request–Response cycle (how browser works)

Day 5-6

DSA: Hashing basics (dict in python/Java/c++)

System Design - Vertical vs Horizontal scaling and Load basics

Day 7

DSA: Practice (mixed 3 easy probs along with 2 medium probs)

System design – Intro to databses (SQL vs NoSQL)

WEEK 2

Day 8-9

DSA: Two pointer technique

System Design – Caching(why,how) and cache examples (Redis concept)

Day 10 -11

DSA: Sliding Window Basics

System Design – Load balancers(basic idea)

Day 12–13

DSA: Stack basics

System Design- API design (REST basics) and HTTP methods (GET, POST)

Day 14

DSA: Practice day (4 problems)

System Design- CDN (Content Delivery Network)

WEEK 3

Day 15–16

DSA: Queue basics

System Design- Database indexing and CAP theorem (basic intuition)

Day 17–18

DSA: Linked List basics

System Design- Sharding vs Replication

Day 19–20

DSA: Recursion basics

System Design- Message queues (Kafka idea)

Day 21

DSA: Practice (mixed)

System Design- Rate limiting (very important)

WEEK 4

Day 22–23

DSA: Sorting basics

System Design- Design URL Shortener (step-by-step)

Day 24–25

DSA: Binary Search

System Design- Design Instagram Feed (basic)

Day 26–27

DSA: Trees intro

System Design- Design Chat System (WhatsApp basic)

Day 28

DSA: Practice problems

System Design- Design File Upload System

Day 29

DSA: Mixed revision

System Design- Revise ALL concepts (make your own notes)

Day 30 🎯

DSA: Solve 3 problems without help

System Design- Mock Design “Design YouTube (basic version)”

PHASE 1 (Days 1–10): “I can code basics”

Learn Python/Java/C++ basics:

- Day 1–2: print, variables, input/output
- Day 3–4: if-else
- Day 5–6: loops (for, while)
- Day 7–8: lists (arrays)
- Day 9–10: functions

Practice:

- sum of numbers
- max in list
- palindrome
- simple patterns

System Design

- What is a website?
- What happens when you open Google?
- What is server?

PHASE 2 (Days 11–20): “I can code a bit”

- Strings
- Arrays
- Hashing (dictionary)
- Basic problem solving (VERY EASY LeetCode)

System Design

- Client–Server
- Database (just idea)
- What is API

PHASE 3 (Days 21–30): “Now things start clicking”

- Arrays problems
- Two pointers (simple)
- Practice daily

System Design

- Caching
- Load balancing
- Try tiny designs like:
 - “How WhatsApp sends a message”
 - “How Instagram stores posts”